

Customer Segmentation & Churn Pattern Analytics in European Banking

A Data-Driven Study of Retail Banking Customer Attrition Across France, Germany, and Spain

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Abstract

Customer churn is one of the most significant hidden costs in retail banking, eroding customer lifetime value and destabilizing revenue. This paper presents a segmentation-driven analysis of 10,000 retail banking customers across France, Germany, and Spain to identify which customer groups are most likely to churn and why. Using exploratory data analysis across geographic, demographic, financial, and engagement dimensions, the study finds an overall churn rate of 20.37%, with pronounced concentration among German customers (32.44%), customers aged 46–60 (51.12%), customers holding three or more banking products (83–100%), and inactive members (26.85% vs. 14.27% for active members). Churned customers also carry 24.26% of total deposit balances, indicating disproportionate value at risk. These findings are translated into segment-specific retention recommendations and a proposed KPI framework for ongoing monitoring via an interactive dashboard.

1. Introduction

1.1 Background and Context

Customer churn represents one of the largest hidden costs in retail banking. When a bank loses an existing customer, it forfeits future revenue streams, absorbs the higher cost of acquiring a replacement customer, and introduces volatility into its revenue base. While most banks already track an aggregate churn rate, this single number rarely answers the questions that matter most for retention strategy: which customer groups are most likely to churn, how churn differs across countries, age groups, and financial profiles, and whether attrition is concentrated among high-value or low-value customers. This project addresses these gaps through systematic, segmentation-driven analytics built on a real customer-level dataset.

1.2 Problem Statement

Despite having rich customer-level data, banks face persistent challenges in identifying high-risk customer segments, understanding how churn differs by geography and demographics, and quantifying the financial profile of customers who leave. Without structured analytics, churn-management strategies remain generic, reactive, and inefficient — retention budgets are spent broadly rather than where they will have the greatest impact.

2. Project Objectives

2.1 Primary Objectives

- Measure the overall customer churn rate across the retail banking portfolio.
- Identify how churn is distributed across customer segments (geography, age, credit profile, tenure, balance).
- Compare churn behaviour across the three European regions represented in the data (France, Germany, Spain).

2.2 Secondary Objectives

- Understand churn concentration among high-value (high-balance, high-salary) customers.
- Evaluate engagement (active membership) and tenure patterns as churn indicators.
- Support strategic retention planning and marketing decision-making with concrete, ranked recommendations.

3. Dataset Description

The analysis uses a customer-level retail banking dataset covering **10,000 customers** across three European markets — France, Germany, and Spain. Each row represents one customer and combines demographic, financial, and behavioural attributes with a binary churn label (*Exited*). Table 1 summarizes the fields used in the analysis.

Column	Description
CustomerId / Surname	Unique customer identifier and surname (removed prior to analysis)
CreditScore	Customer creditworthiness score
Geography	Customer's country: France, Spain, or Germany
Gender	Male / Female
Age	Customer age in years
Tenure	Number of years the customer has held an account with the bank
Balance	Current account balance
NumOfProducts	Number of bank products held by the customer
HasCrCard	Whether the customer holds a credit card (1 = yes)
IsActiveMember	Engagement indicator (1 = active)
EstimatedSalary	Estimated annual salary
Exited	Churn indicator and target variable (1 = churned)

Table 1. Dataset schema used for segmentation and churn analysis.

4. Analytical Methodology

4.1 Data Ingestion, Validation & Cleaning

The dataset was loaded and validated to confirm consistency of binary fields (*HasCrCard*, *IsActiveMember*, *Exited*) and accuracy of churn labelling. Non-analytical identifier fields (*Surname*) were removed, and categorical fields were prepared for group-wise aggregation. Derived segmentation fields (age band, credit-score band, balance segment) were engineered to support the analysis described below.

4.2 Customer Segmentation Design

Five segmentation dimensions were constructed to slice the churn signal:

- **Geographic segmentation** — France, Spain, Germany.
- **Age segmentation** — Under 30, 30–45, 46–60, 60+.
- **Credit score bands** — Low, Medium, High.
- **Tenure groups** — New, mid-term, and long-term customers (0–10 years).
- **Balance segments** — Zero-balance, low-balance, and high-balance customers.

Churn distribution was then examined at the overall level, within each segment, and across segment interactions (e.g., geography × gender), followed by a targeted deep-dive into high-value customers to quantify the revenue at risk from attrition.

5. Exploratory Data Analysis & Results

5.1 Overall Churn Rate

Of the 10,000 customers in the portfolio, **2,037 have churned**, giving an overall churn rate of **20.37%**. This baseline is the reference point against which every segment below is compared: a segment churning meaningfully above 20.37% is a retention priority, while a segment churning well below it represents a comparatively stable base.

5.2 Geographic Churn Patterns

Churn is heavily concentrated in Germany. Although Germany contributes only 2,509 of the 10,000 customers (25.1% of the base), its churn rate of **32.44%** is roughly double that of France (16.15%) and Spain (16.67%). This is a critical finding: geographic risk is not proportional to customer volume — Germany is a comparatively small market generating an outsized share of attrition.

Geography	Customers	Churned	Churn Rate
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France	5,014	810	16.15%
Spain	2,477	413	16.67%
Germany	2,509	814	32.44%

Table 2. Churn rate by geography.

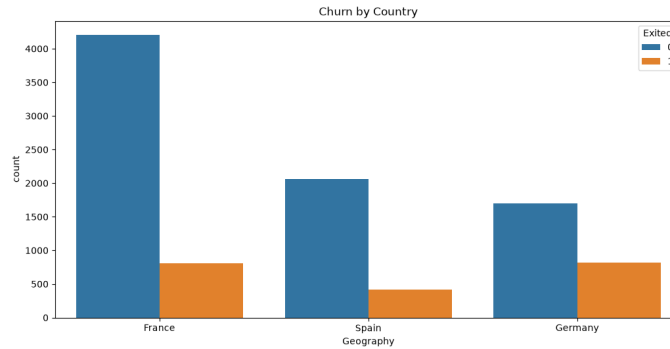


Figure 1. Distribution of retained (0) vs. churned (1) customers by country.

5.3 Gender-Based Churn Differences

Female customers churn at nearly one and a half times the rate of male customers: **25.07%** versus **16.46%**. This gap holds directionally across all three countries, but is widest in Germany, where female churn reaches 37.55% compared to 27.81% for male customers (Table 3).

Geography	Female Churn Rate	Male Churn Rate
France	20.34%	12.71%
Germany	37.55%	27.81%
Spain	21.21%	13.11%

Table 3. Gender churn rate by geography (interaction effect).

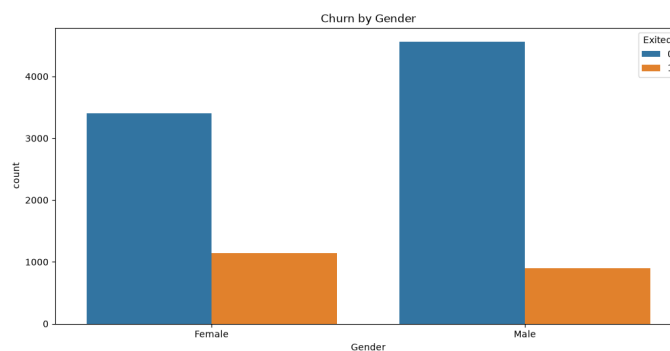


Figure 2. Retained vs. churned customers by gender across the full portfolio.

5.4 Age and Life-Stage Churn

Age is the single strongest demographic driver of churn in this dataset. Customers aged **46–60 churn at 51.12%** — more than double the portfolio baseline — while customers under 30 churn at only 7.52%. The boxplot in Figure 3 confirms this pattern: churned customers skew markedly older, with a median age around 45 versus roughly 36 for retained customers.

Age Group	Customers	Churned	Churn Rate
Under 30	1,968	148	7.52%
30–45	5,921	932	15.74%
46–60	1,647	842	51.12%

60+	464	115	24.78%
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Table 4. Churn rate by age segment.

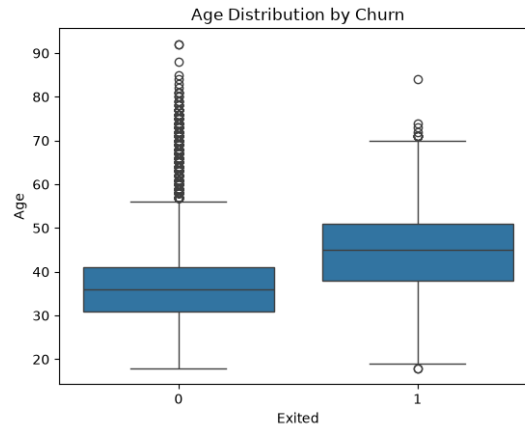


Figure 3. Age distribution of retained (0) vs. churned (1) customers.

5.5 Tenure vs. Churn

Unlike age and geography, tenure shows a comparatively flat relationship with churn: rates hover between 17% and 23% across all tenure years from 0 to 10, with no strong monotonic trend (Figure 4). The slight elevation at year 0 (23.00%) suggests a modest early-relationship risk window, but tenure alone is a weak predictor of attrition compared to age, geography, or product holding.

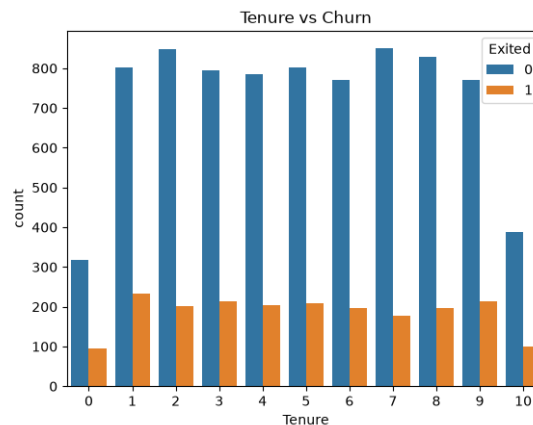


Figure 4. Customer count and churn split by tenure (years with the bank).

5.6 Product Holding vs. Churn

The number of banking products held is the most dramatic churn signal in the dataset. Customers with a single product churn at 27.71%, and those with two products churn at just 7.58% — the healthiest segment in the portfolio. However, customers holding **three products churn at 82.71%**, and every single customer holding **four products has churned (100%)**. This inverted pattern strongly suggests that 3–4 product bundles are being reached through forced cross-selling, product mismatch, or as a late-stage retention offer that ultimately fails, rather than reflecting genuine multi-product loyalty.

Products Held	Customers	Churned	Churn Rate
1	5,084	1,409	27.71%
2	4,590	348	7.58%
3	266	220	82.71%
4	60	60	100.00%

Table 5. Churn rate by number of products held.

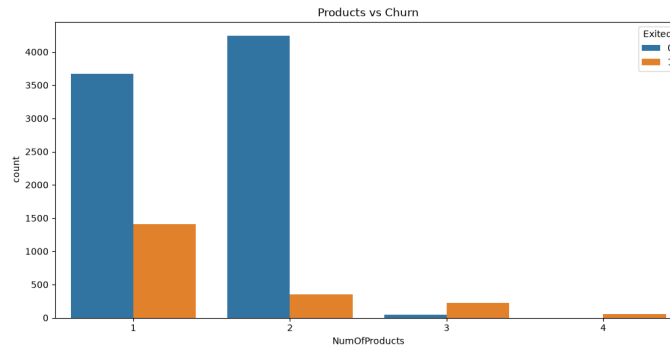


Figure 5. Retained vs. churned customers by number of products held.

5.7 Engagement (Active Membership) vs. Churn

Active members churn at 14.27%, while inactive members churn at 26.85% — nearly double. Engagement status is therefore a strong, actionable, and comparatively simple signal: banks can identify inactive customers well before they formally exit and intervene with targeted re-engagement.

5.8 Correlation Summary

A correlation check against the churn label corroborates the segment-level findings: age has the strongest positive association with churn ($r = 0.29$), followed by balance ($r = 0.12$), while active membership shows the strongest negative association ($r = -0.16$). Credit score, tenure, product count (as a linear term), salary, and credit-card ownership show weak or negligible linear correlation individually — their predictive value only emerges once they are segmented into bands, as in Sections 5.4–5.6.

6. High-Value Customer Churn Analysis

Balance-based segmentation reveals that churn risk rises with account value. Zero-balance customers churn at only 13.82%, low-balance customers at 20.58%, and high-balance customers at 25.23% — meaningfully above the portfolio average (Table 6).

Balance Segment	Customers	Churned	Churn Rate
Zero Balance	3,617	500	13.82%
Low Balance	1,584	326	20.58%
High Balance	4,799	1,211	25.23%

Table 6. Churn rate by account balance segment.

Restricting the population to the top quartile by balance (accounts $\geq \text{€}127,644$) yields 2,500 high-value customers churning at **23.68%**. Narrowing further to a “premium” segment (top-quartile balance *and* above-median estimated salary, $n = 1,254$) produces a churn rate of **24.40%**. Both figures sit above the portfolio baseline, confirming that the bank is not simply losing low-value, disengaged accounts — it is losing a disproportionate share of its most valuable relationships.

In direct revenue terms, churned customers collectively hold **€185.6 million** in balances, equivalent to **24.26% of total deposits** held across the entire 10,000-customer portfolio — a share almost exactly proportional to their 20.37% share of customers, indicating that high-value churners are not statistical outliers but a systematic pattern requiring dedicated relationship management. Notably, the combination of high balance and inactivity is especially dangerous: high-value customers who are also inactive churn at **30.47%**, the highest risk combination identified in this study.

This risk is compounded by geography: Germany carries the highest average balance per customer (€119,730) of any market, with **0% of German customers holding a zero balance**, versus roughly 48% zero-balance customers in both France and Spain. In other words, Germany is simultaneously the highest-churn *and* highest-balance-per-customer market, making it the single largest source of revenue-at-risk in the portfolio and the clearest priority for retention investment.

7. Key Performance Indicators

The following KPI framework, computed directly from the dataset, is proposed for ongoing monitoring via the companion dashboard described in Section 8.

KPI	Definition	Computed Value
Overall Churn Rate	% of customers who exited	20.37%
Segment Churn Rate (range)	Churn % across defined segments	7.52% – 51.12% (age bands)
High-Value Churn Ratio	Churn rate among top-quartile balance customers	23.68%
Geographic Risk Index	Highest vs. lowest regional churn rate	Germany 32.44% vs. France 16.15%
Engagement Drop Indicator	Churn rate: inactive vs. active members	26.85% vs. 14.27%

Table 7. Key performance indicator framework and current values.

8. Proposed Streamlit Dashboard

To operationalize these findings, a Streamlit web application is proposed with four core modules: an overall churn summary, a geography-wise churn visualization, an age-and-tenure churn comparison view, and a high-value customer churn explorer. Users would be able to apply segment filters, view KPIs update dynamically, and drill down from portfolio-level metrics into individual segment detail — turning the static findings in this paper into a live, continuously updated retention-monitoring tool for stakeholders.

9. Discussion and Recommendations

The analysis points to five concrete, ranked actions for retention strategy:

- **Prioritize Germany.** Despite representing only a quarter of customers, Germany drives disproportionate churn and holds the highest average balance per customer. A dedicated German retention program (proactive relationship-manager outreach, localized service review) offers the highest expected return of any regional intervention.
- **Target the 46–60 age band.** With a churn rate above 51%, this life-stage segment (likely approaching retirement-related financial decisions) warrants tailored product review and dedicated advisory contact rather than generic marketing.
- **Investigate the 3–4 product bundle.** Near-total churn among customers with three or more products suggests these bundles are being mis-sold, forced, or offered too late in the relationship. A qualitative review of how these products are sold is recommended before any further cross-sell push.
- **Build an engagement early-warning system.** Because inactive members churn at nearly double the rate of active members, flagging inactivity (e.g., 60–90 days without a transaction) can trigger a re-engagement workflow before the customer formally exits.
- **Protect high-value relationships.** High-balance and premium customers churn above the portfolio average and represent a disproportionate share of deposits at risk; these accounts should be routed to senior relationship managers rather than standard retention scripts.

10. Conclusion

This project demonstrates that a single aggregate churn rate of 20.37% conceals sharply different risk levels across customer segments — ranging from 7.52% among younger customers to over 51% among customers aged 46–60, and from near-zero product-driven churn among two-product holders to near-total churn among three-and four-product holders. Geography, age, product holding, engagement status, and account balance each independently sharpen the picture of who is at risk and why. By translating these patterns into a structured KPI framework and a proposed live dashboard, this analysis equips decision-makers with the segmentation-driven, actionable insight needed to move from reactive, blanket retention spending toward targeted, data-driven strategies focused on Germany, the 46–60 age band, over-bundled product holders, disengaged members, and high-value accounts.

Tools & Data

Analysis performed in Python (pandas, seaborn, matplotlib) on a 10,000-record European retail banking customer dataset (France, Germany, Spain). Proposed dashboard delivery via Streamlit. Prepared as part of a project portfolio in collaboration with Unified Mentor.